Zürcher Hochschule
für Angewandte Wissenschaften

School of
Management and Law

Agent-Based Credit Risk Modeling

{'Summer School Bayonne 2026'}

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Program for today

Program: Agent-Based Credit Risk Modeling Workshop
- Introduction, organization of groups, installation of software, brainstorming
- Working in groups on the tasks; expert sessions
- Presentation of results

*breaks are included as needed

Workshop procedure and philosophy

Image Credit: https://revistaempresarial.com

- No continuous presentations by the lecturer
- Group work (2 students per group)
- The groups are mixed (BSc/MSc OR beginners/professionals)
- Each "expert group" is responsible for one "topic“
- Methods are explained in detail in "technical sessions“
- Students ask the members of the "expert groups" for help
- The more experienced students help the others
- At the end of the day, the groups present their results
- The presentations are graded pass/fail.

Prerequisites

Software:

- GitHub Codespaces
  - Python 3.12
  - Visual Studio Code

Course GitHub Repository (create a fork in your personal GitHub account)

- https://github.com/mario-gellrich-zhaw/summerschool_2026

Mandatory reading

- https://mesa.readthedocs.io/en/stable/overview.html

Credit Risk

"Credit risk, counterparty risk or default risk is a term used in the banking industry to refer to the risk that a borrower is unable or unwilling to repay the loans granted to it in full or in accordance with the contract. In general, credit risk is the most significant type of risk for credit institutions."

Source: Wikipedia

Keyword pinboard (basis to define group tasks)

(Topic 1) Introduction credit risk modeling (terms & concepts)
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(Topic 2) Loan calculation example (use-case)
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(Topic 3) Integration of foundation models (FM)
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(Topic 4) Scheduling agents in an agent-based model
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Keyword pinboard (basis to define group tasks)

(Topic 5) Lender & Borrower classes (sketch)

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(Topic 6) CreditModel class (sketch)

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(Topic 7) Agent-based simulation (input & expected output)

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(Topic 8) Sensitivity analysis (input & expected output)

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Structure of the final presentation (HTML, part of the WebApp)

The presentation (provided by every group at the end of the day) must have the following structure.

1. Introduction

1.1 Background

1.2 Problem

1.3 Objectives

1.4 Research Question

2. Materials and Methods

2.1 Credit risk modeling – terms and concepts

2.2 Pythons mesa library for agent-based modeling

2.3 Agent-based credit risk model classes, methods, parameters

2.4 Integration of a Foundation Model (FM)

2.5 Agent-based credit risk model simulations

2.6 Sensitivity analysis

2.7 Web Application

3. Results & Discussion

4. Conclusions